

LINEAR ALGEBRA - UE24MA241B

Mini Project

Semester-4 Section-I

Title- Traffic Volume Prediction and Pattern Analysis using Linear Algebra

Team Members-

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Problem Statement-

Traffic congestion is a major issue in urban areas, influenced by multiple factors such as weather conditions, time, and environmental variables. Understanding and predicting traffic patterns can help in better planning and congestion management.

In this project, we apply concepts from linear algebra to analyze a real-world traffic dataset. The goal is to transform raw data into a structured mathematical form, reduce unnecessary information, and use mathematical techniques to predict traffic volume and identify patterns.

Link to Dataset- [Interstate traffic data](#)

The dataset used is the Metro Interstate Traffic Volume dataset from Kaggle.

Features include:

- Temperature
- Rainfall (rain_1h)
- Snowfall (snow_1h)
- Cloud coverage (clouds_all)
- Traffic volume (target variable)

Each row represents traffic conditions at a specific time.

Linear Algebra Flow Applied-

Step-1 Matrix Description

The dataset is converted into matrix form:

- Matrix A features (temperature, rain, snow, clouds)
- Vector b traffic volume

This forms a system: $Ax=b$

- Purpose
To organize multiple influencing factors into a single mathematical structure.
- Impact
This allows us to study how different factors jointly affect traffic volume, instead of analyzing them separately.

Step-2 Matrix Simplification

RREF / Gaussian Elimination

The matrix is simplified using row operations to identify independent and dependent variables.

- Purpose
To remove redundant or repetitive features.
- Impact
Eliminates unnecessary factors, improves efficiency, ensures only meaningful variables influence predictions

```
[5] ✓ 0.0s Pyth
... Pivot columns (0-indexed): (0, 1, 4, 5)
Rank of sub-matrix : 4

RREF (first 10 rows):
Matrix([[1, 0, 0, 0, 0, 0, -0.457821476533691, -2.71075200824588], [0, 1, 0, 0, 0, 0, 0, 0], [0, 0, 0, 0, 1, 0, 0, 0], [0, 0, 0, 0, 0, 1, 0, 0], [0, 0, 0, 0, 0, 0, 1, 0], [0, 0, 0, 0, 0, 0, 0, 1], [0, 0, 0, 0, 0, 0, 0, 0], [0, 0, 0, 0, 0, 0, 0, 0], [0, 0, 0, 0, 0, 0, 0, 0], [0, 0, 0, 0, 0, 0, 0, 0]])
Contains NaN: False
Contains Inf: False
```

Step-3 Structure of the Space

Analysis of:

- Column space
- Row space
- Rank of the matrix

Purpose

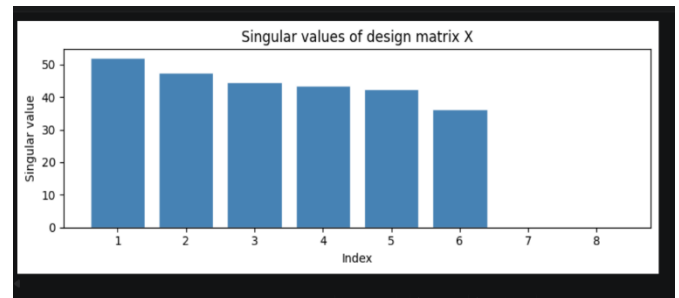
To determine how many independent factors actually influence traffic.

Impact

- Identifies key variables
- Simplifies the model
- Avoids overcomplication

```
[7] ✓ 0.0s
... Matrix X : 2000 rows x 8 cols
Rank      : 6
Nullity    : 2
Rank + Nullity = 6 + 2 = 8 = n_cols ✓

Null space vectors: 2
```



Step-4 Removal of Redundancy (Basis Selection)

A basis is selected from the feature set to retain only independent vectors.

Purpose

To eliminate duplicate or highly correlated features.

Impact

- Reduces noise
- Prevents overcounting of similar effects
- Improves prediction clarity

```
✓ 0.0s
All columns      : ['bias', 'temp', 'rain_1h', 'snow_1h', 'clouds_all', 'hour', 'day_of_week', 'month']
Independent cols : ['bias', 'temp', 'clouds_all', 'hour', 'day_of_week', 'month']
Dropped cols     : ['rain_1h', 'snow_1h']

Reduced X shape: (2000, 6)
```

Step-5 Orthogonalization (Gram–Schmidt Process)

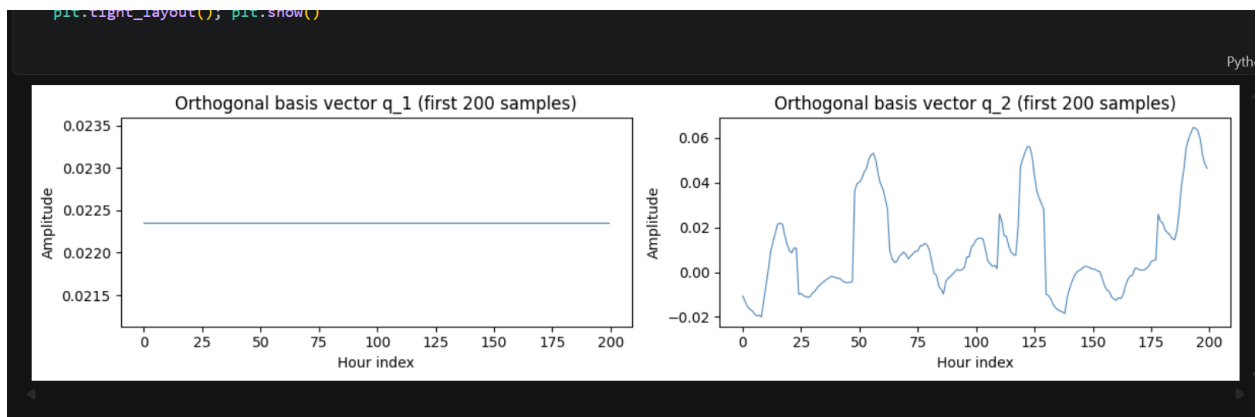
The basis vectors are converted into orthogonal vectors.

Purpose

To ensure each feature contributes independently without overlap.

Impact

- Separates the effect of each factor
- Improves numerical stability
- Makes predictions more reliable



Step-6 Projection

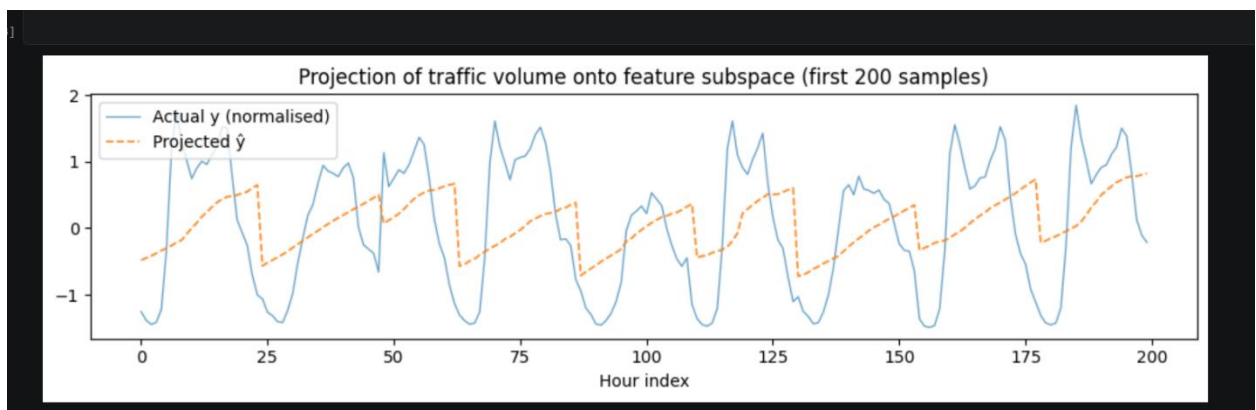
Data is projected onto the feature subspace.

Purpose

To estimate traffic values when exact solutions are not possible.

Impact

- Handles real-world inconsistencies
- Provides closest possible approximation



Step-7 Prediction using Least Squares

The least squares method is used:

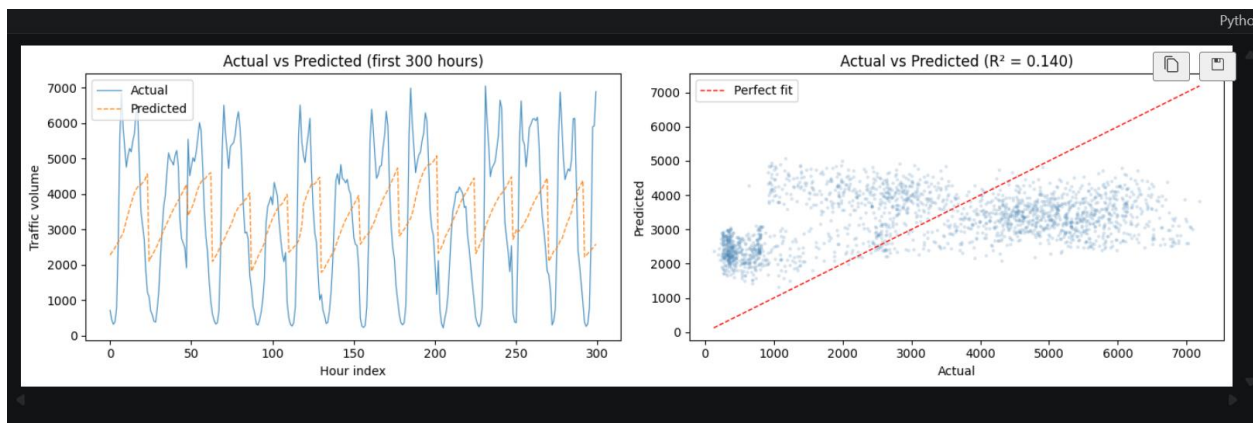
$$\hat{x} = (A^T A)^{-1} A^T b$$

Purpose

To minimize the difference between actual and predicted traffic values.

Impact

- Produces accurate traffic predictions
- Reduces overall prediction error
- Core method for solving real-world data problems



Step-8 Pattern Discovery using Eigenvalues

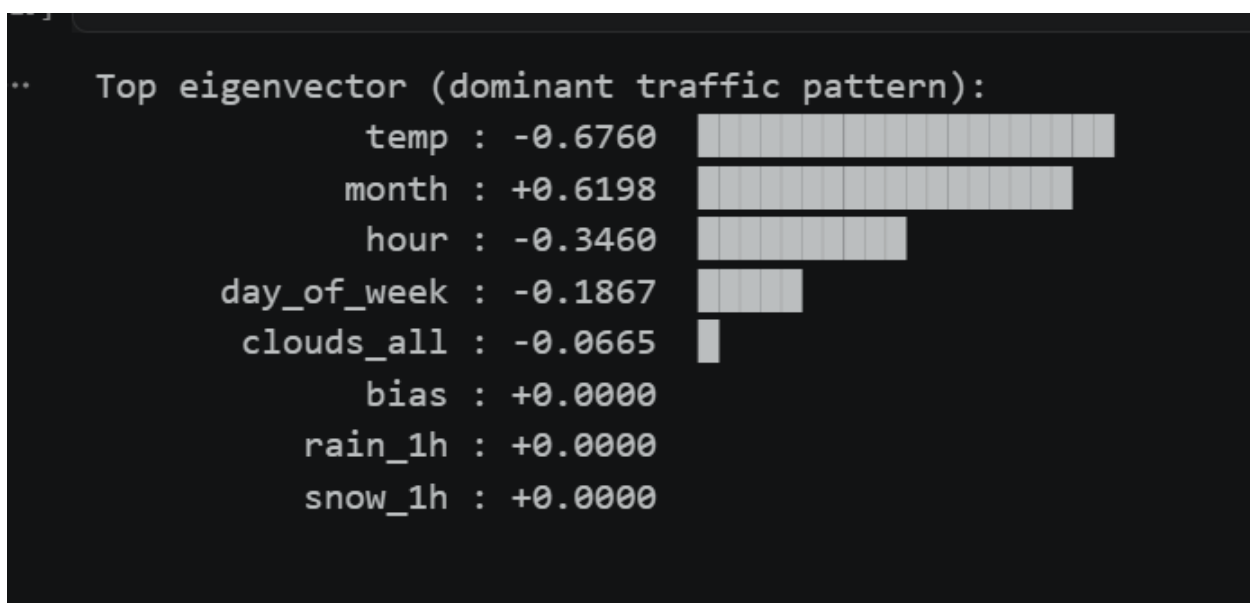
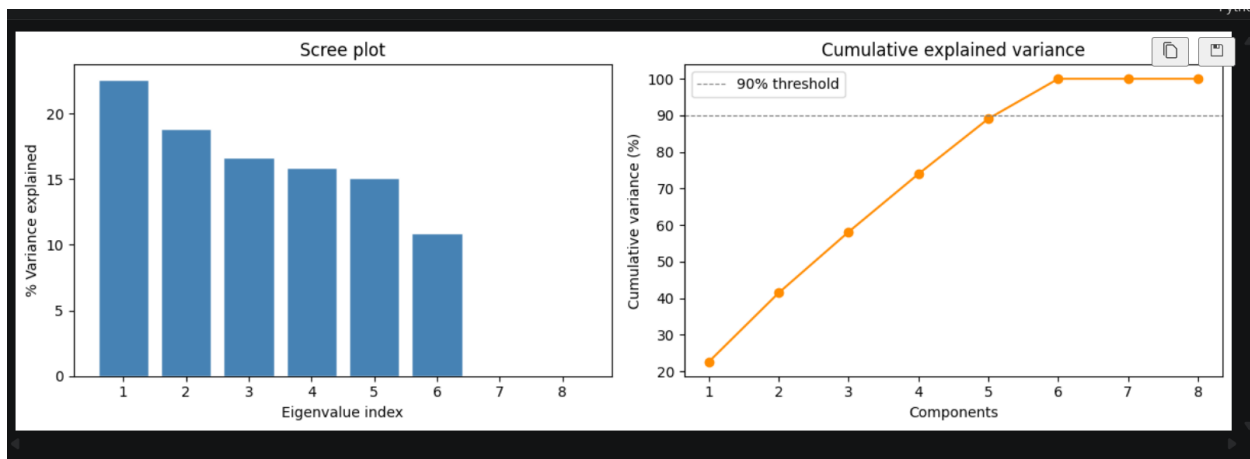
Eigenvalues and eigenvectors are computed from the data.

Purpose

To identify dominant patterns in traffic behavior.

Impact

- Detects peak traffic trends
- Identifies strong influencing factors
- Reveals hidden patterns in data



Final Output

The final output includes:

- Predicted traffic volume
- Comparison of actual vs predicted values
- Identified traffic patterns such as:
 - Peak hours
 - Weather influence

Impact

These results can be used for:

- Traffic management
- Route optimization
- Smart city planning

